

# **Anatomy & Physiology of the Reproductive System**

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## **Learning Objectives**

Upon completion of the lecture, you will be able to:

1. Define the key terms used in this lecture.
2. Describe changes of puberty in males and females.
3. Contrast the structure and function of the major external and internal female genital organs.
4. Outline the phases of the menstrual cycle, the dominant hormones involved, and the changes taking place in each phase.
5. Classify external and internal male reproductive structures and the function of each in hormonal regulation

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## **KEY TERMS**

Breasts, cervix, endometrium, estrogen, fallopian tubes, follicle-stimulating hormone (FSH), luteinizing hormone (LH), menarche, menstruation, ovaries, ovulation, penis, progesterone, testes, uterus, vagina, and vulva.

## **OVERVIEW**

The reproductive system consists of organs that function in the production of offspring.

Nurses need to have a thorough understanding of the anatomy and physiology of the male and female reproductive systems to be able to assess the health of these systems, to promote reproductive system health, to care for conditions that might affect the reproductive organs, and to provide client teaching concerning the reproductive system.

This lecture reviews the female and male reproductive systems and the menstrual cycle as it relates to reproduction.

## Puberty

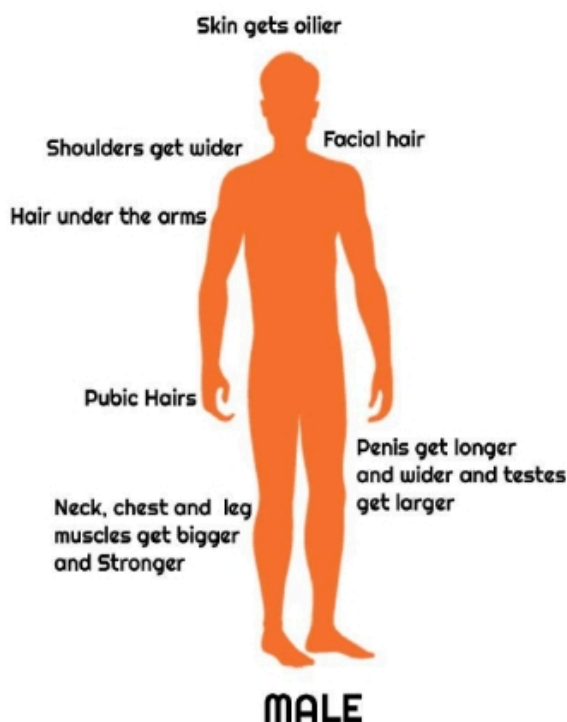
Puberty is a period of rapid change in the lives of boys and girls during which the reproductive systems mature and become capable of reproduction. Puberty ends when mature sperm are formed or when regular menstrual cycles occur.

**In girls**, the first menstrual period (menarche) occurs 2 to 2 years later (age 11 to 15 years), and pubertal changes typically occur as:

1. Growth spurt
2. Increase in the transverse diameter of the pelvis
3. Breast development
4. Growth of pubic hair
5. Onset of menstruation
6. Growth of axillary hair
7. Vaginal secretions

**In boys**, male hormonal changes normally begin between 10 and 16 years of age, and pubertal changes typically occur as:

1. Increase in weight
2. Growth of testes
3. Growth of face, axillary, and pubic hair
4. Voice changes
5. Penile growth
6. Increase in height
7. Spermatogenesis (production of sperm)

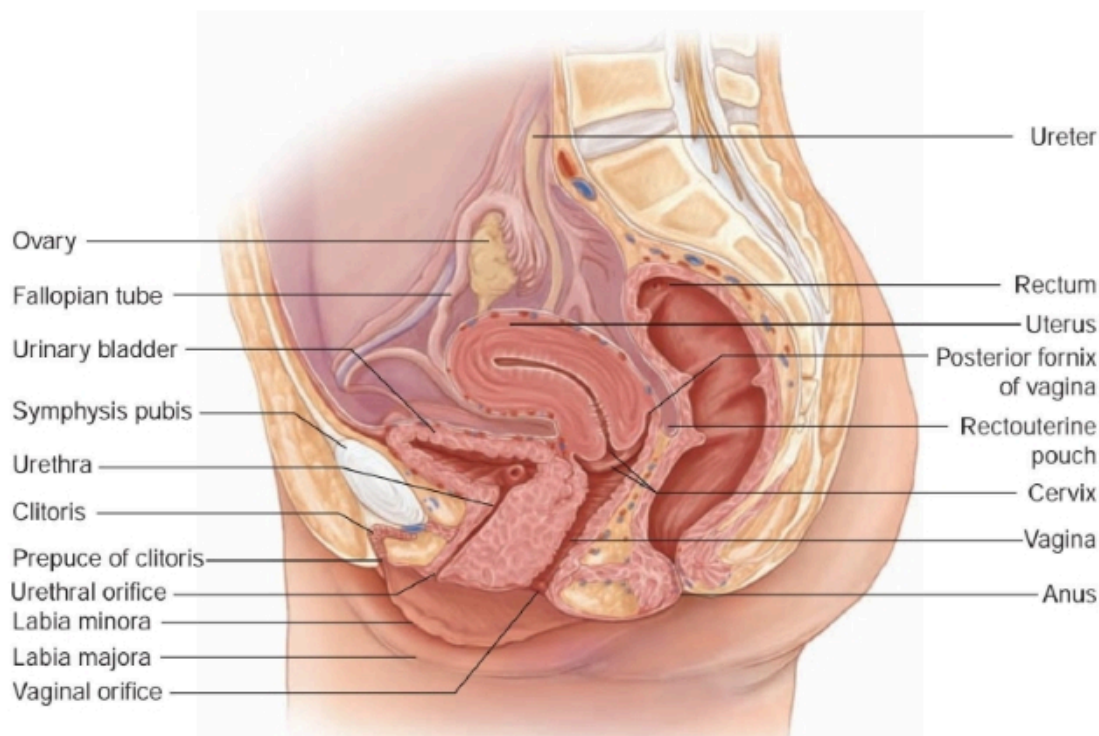


## Female Reproductive System

The female reproductive system consists of external genitalia, internal genitalia, and accessory structures such as the mammary glands (breasts).

### External Genitalia

The female external genitalia are collectively called the vulva. They include the mons pubis, labia majora, labia minora, fourchette, clitoris, vaginal vestibule, and perineum.



### Mons pubis

The mons pubis (mons veneris) is a pad of fatty tissue covered by coarse skin and hair. It protects the symphysis pubis and contributes to the rounded contour of the female body.

### Labia majora

The labia majora are two folds of fatty tissue on each side of the vaginal vestibule. Many small glands are located on the moist interior surface.

### Labia minora

The labia minora are two thin, soft folds of tissue that are seen when the labia majora are separated. Secretions from sebaceous glands in the labia are bactericidal to reduce infection and lubricate and protect the skin of the vulva.



## Fourchette

The fourchette is a fold of tissue just below the vagina, where the labia majora and the labia minora meet. It is also known as the obstetrical perineum.

## Clitoris

The clitoris is a small, erectile body in the most anterior portion of the labia minora. It is similar in structure to the penis. Functionally, it is the most erotic, sensitive part of the female genitalia.

## Vaginal vestibule

The vaginal vestibule is the area seen when the labia minora are separated and includes five structures:

1. **The urethral meatus** lies approximately 2 cm below the clitoris. It has a foldlike appearance with a slit type of opening, and it serves as the exit for urine.
2. **Skene ducts (paraurethral ducts)** are located on each side of the urethra and provide lubrication for the urethra and the vaginal orifice.
3. **The vaginal introitus** is the division between the external and internal female genitalia.
4. **The hymen** is a thin elastic membrane that closes the vagina from the vestibule to various degrees.
5. **The ducts of the Bartholin glands (vulvovaginal glands)** provide lubrication for the vaginal introitus during sexual arousal and are normally not visible.

## Perineum

The perineum is a strong, muscular area between the vaginal opening and the anus. The elastic fibers and connective tissue of the perineum allow stretching to permit the birth of the fetus.

## Internal Genitalia

The internal genitalia are the vagina, uterus, fallopian tubes, and ovaries.

## Vagina

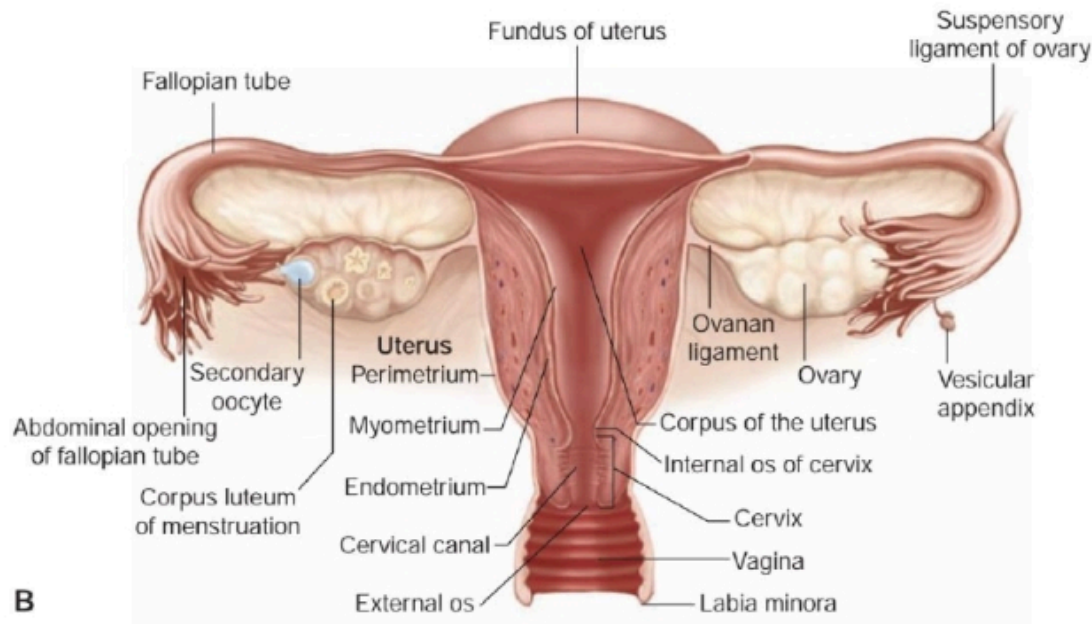
The vagina is a tubular structure made of muscle and membranous tissue that connects the external genitalia to the uterus. An average adult vagina is slightly curved and can range between 7 to 12 cm in length, but everybody is different. The vagina is self-cleansing and during the reproductive years maintains a normal acidic pH of 4 to 5.

### The vagina has three functions:

1. Provides a passageway for sperm to enter the uterus.
2. Allows drainage of menstrual fluids and other secretions.
3. Provides a passageway for the infant's birth.

## Uterus

The uterus (womb) is a hollow muscular organ in which a fertilized ovum is implanted, an embryo forms, and a fetus develops. It is shaped like an upside-down pear or light bulb. In a mature, nonpregnant woman, it weighs approximately 60 g and is 7.5 cm long, 5 cm wide, and 1 to 2.5 cm thick. The uterus lies between the bladder and the rectum above the vagina.



The uterus is separated into three parts: fundus, corpus, and cervix. The fundus (upper part) is broad and flat. The fallopian tubes enter the uterus on each side of the fundus. The corpus (body) is the middle portion, and it plays an active role in menstruation and pregnancy.

### The fundus and the corpus have three distinct layers:

1. **The perimetrium** is the outermost or serosal layer that envelops the uterus.
2. **The myometrium** is the middle muscular layer that functions during pregnancy and birth.
3. **The endometrium** is the inner or mucosal layer that is functional during menstruation and implantation of the fertilized ovum. It is governed by cyclical hormonal changes.

The **cervix** (lower part) is narrow and tubular and opens into the upper vagina. The cervix consists of a cervical canal with an internal opening near the uterine corpus (internal os) and an opening into the vagina (the external os).

### The mucosal lining of the cervix has four functions:

1. Lubricates the vagina.
2. Acts as a bacteriostatic agent.
3. Provides an alkaline environment to shelter deposited sperm from the acidic pH of the vagina.
4. Produces a mucous plug in the cervical canal during pregnancy.

## Fallopian tubes

The fallopian tubes, also called uterine tubes or oviducts, extend laterally from the uterus, one to each ovary. They vary in length from 8 to 13.5 cm.

### Each tube has four sections:

1. **The interstitial portion** extends into the uterine cavity and lies within the wall of the uterus.
2. **The isthmus** is a narrow area near the uterus.
3. **The ampulla** is the wider area of the tube and is the usual site of fertilization.
4. **The infundibulum** is the funnel-like enlarged distal end of the tube.

Fingerlike projections from the infundibulum, called **fimbriae**, hover over each ovary and “capture” the ovum (egg) as it is released by the ovary at ovulation.

### The four functions of the fallopian tubes are to provide the following:

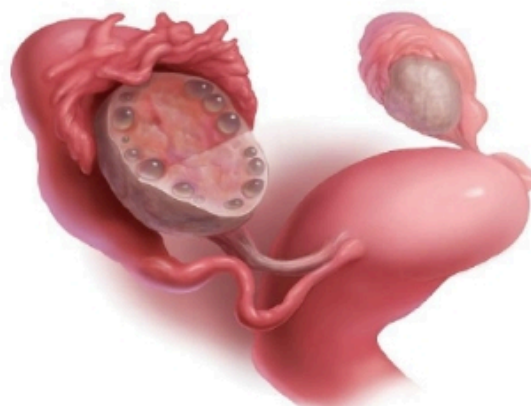
1. A passageway in which sperm meet the ovum.
2. The site of fertilization (usually the outer one-third of the tube).
3. A safe, nourishing environment for the ovum or zygote (fertilized ovum).
4. The means of transporting the ovum or zygote to the corpus of the uterus.

## Ovaries

The ovaries are two almond-shaped glands, each about the size of a walnut. They are located in the lower abdominal cavity, one on each side of the uterus, and are held in place by ovarian and uterine ligaments.

### The ovaries have two functions:

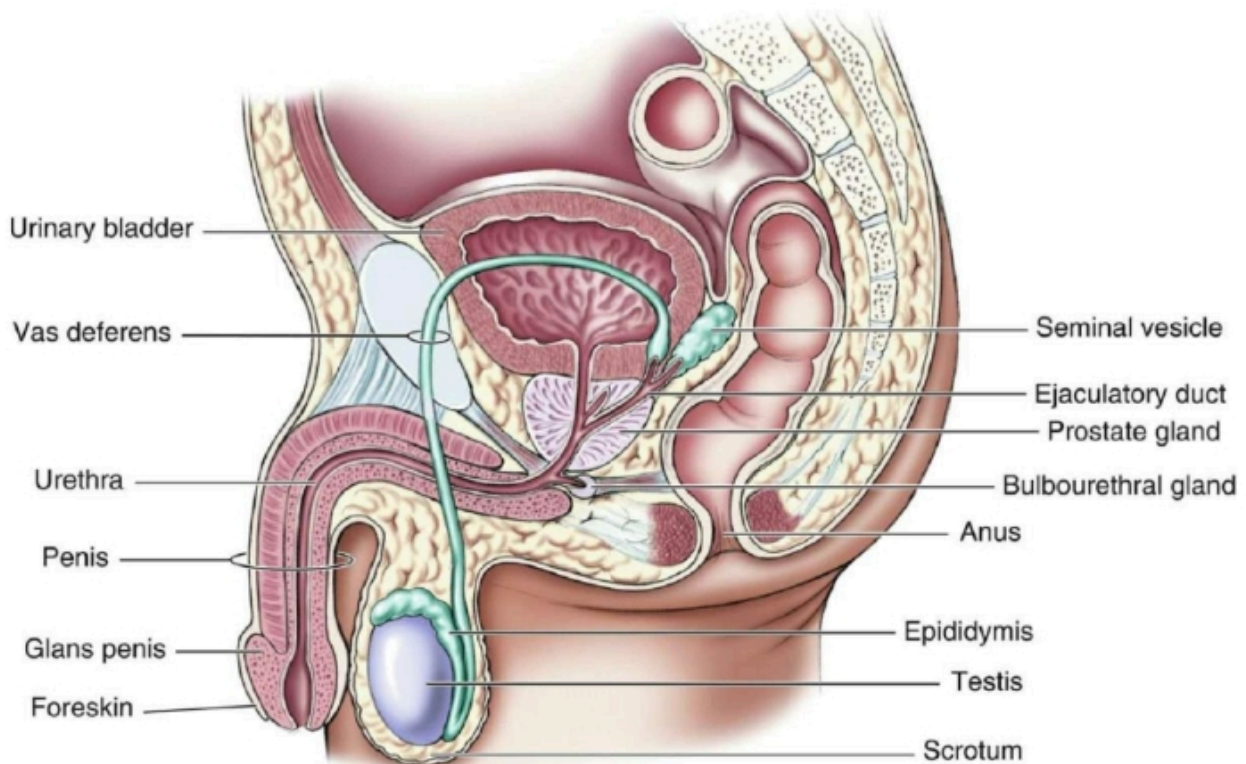
1. Production of hormones, chiefly estrogen and progesterone.
2. Stimulation of an ovum's maturation during each menstrual cycle.





## Male Reproductive System

The male reproductive system consists of external and internal organs.



### External Genitalia

The penis and the scrotum, which contains the testes, are the male external genitalia.

#### Penis

The penis consists of the glans and the body. The glans is the rounded, distal end of the penis. It is visible on a circumcised penis but is hidden by the foreskin on an uncircumcised penis. At the tip of the glans is an opening called the urethral meatus. The body of the penis contains the urethra (the passageway for sperm and urine) and erectile tissue (the corpus spongiosum and two corpora cavernosa).

#### The penis has two functions:

1. Provides a duct to expel urine from the bladder.
2. Deposits sperm in a woman's vagina to fertilize an ovum.

#### Scrotum

The scrotum is a sac that contains the testes. The scrotum is suspended from the perineum, keeping the testes away from the body and thereby lowering their temperature, which is necessary for normal sperm production (spermatogenesis).

## Internal Genitalia

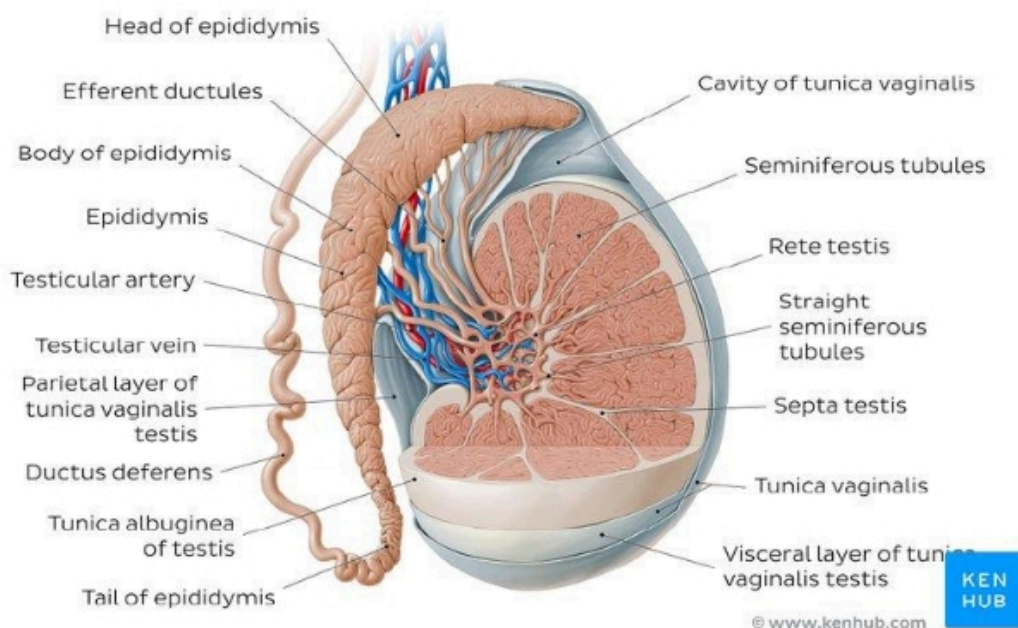
The internal genitalia include the testes, vas deferens, prostate, seminal vesicles, ejaculatory ducts, urethra, and accessory glands.

### Testes

The testes (testicles) are a pair of oval glands housed in the scrotum.

**They have two functions:**

1. Manufacture male germ cells (spermatozoa or sperm).
2. Secrete male hormones (androgens).



Sperm are made in the convoluted seminiferous tubules that are contained within the testes. Sperm production begins at puberty and continues throughout the life span of the male. The production of testosterone, the most abundant male sex hormone, begins with the anterior pituitary gland. Under the direction of the hypothalamus, the anterior pituitary gland secretes follicle-stimulating hormone (FSH) and luteinizing hormone (LH). FSH and LH initiate the production of testosterone in the Leydig cells of the testes.

### Testosterone Functions:

1. Increases muscle mass and strength.
2. Promotes growth of long bones.
3. Increases basal metabolic rate.
4. Enhances production of red blood cells.
5. Produces enlargement of vocal cords.
6. Affects the distribution of body hair.



## Ducts

Each epididymis, one from each testicle, stores the sperm. The sperm may remain in the epididymis for **2 to 10 days**, during which time they mature and then move on to the vas deferens.

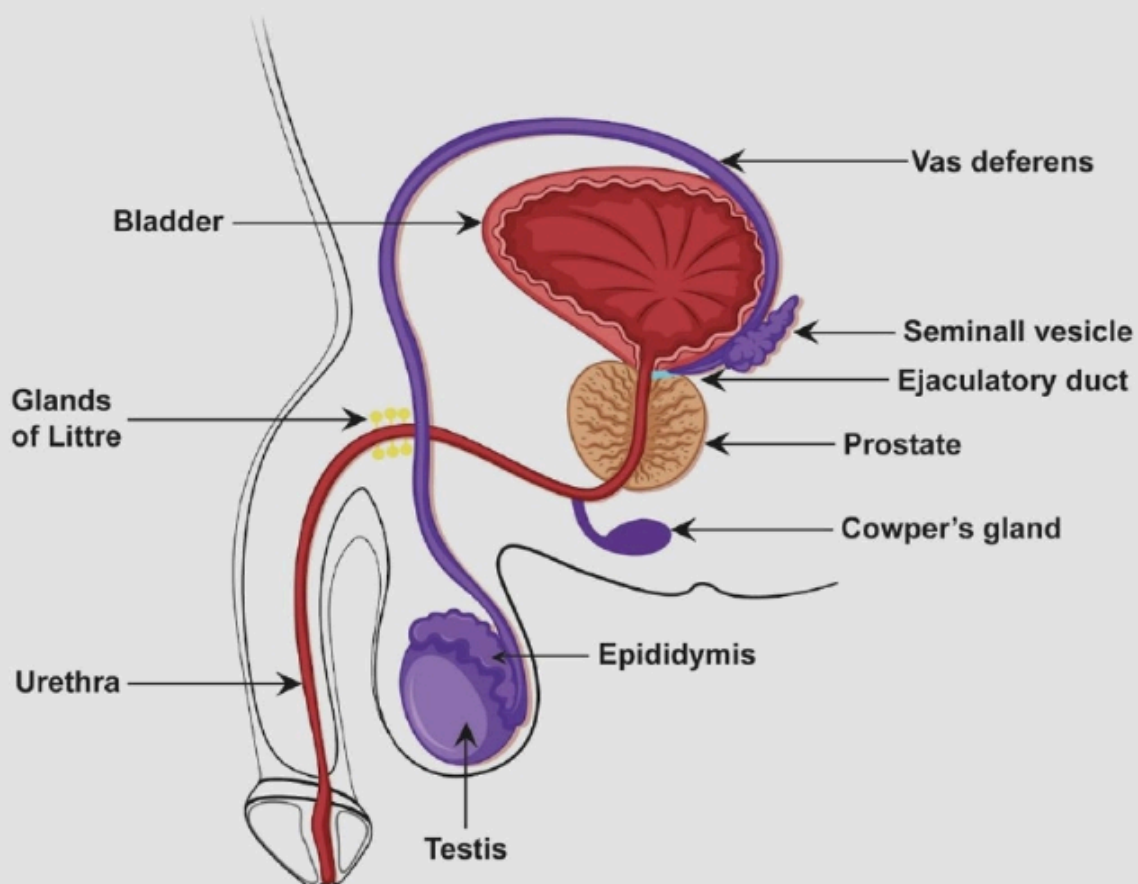
The ejaculatory ducts then enter the back of the prostate gland and connect to the upper part of the urethra, which is in the penis. The urethra transports both urine from the bladder and semen from the prostate gland to the outside of the body, although not at the same time.

## Accessory glands

The accessory glands are the seminal vesicles, the prostate gland, and the bulbourethral glands, also called Cowper's glands.

**The accessory glands have three functions:**

1. Nourish the sperm.
2. Protect the sperm from the acidic environment of the woman's vagina.
3. Enhance the motility (movement) of the sperm.
4. The combined seminal plasma and sperm are called semen. Semen may be secreted during sexual intercourse before ejaculation.



## Reproductive Cycle and Menstruation

### Stages of the Menstrual Cycle Includes:

- A. The ovarian cycle consists of three phases: **follicular (preovulatory) phase, ovulation, and luteal (postovulatory) phase.**
- B. The uterine cycle has three phases: **menses, proliferative and secretory.**

The female reproductive cycle consists of regular changes in secretions of the anterior pituitary gland, the ovary, and the endometrial lining of the uterus. The anterior pituitary gland, in response to the hypothalamus, secretes follicle-stimulating hormone (FSH) and luteinizing hormone (LH). FSH stimulates maturation of a follicle in the ovary that contains a single ovum.

Several follicles start maturing during each cycle, but usually only one reaches final maturity. The maturing ovum and the corpus luteum (the follicle left empty after the ovum is released) produce increasing amounts of estrogen and progesterone, which leads to enlargement of the endometrium.

A surge in LH stimulates final maturation and the release of an ovum. Approximately 2 days before ovulation, the vaginal secretions increase noticeably.

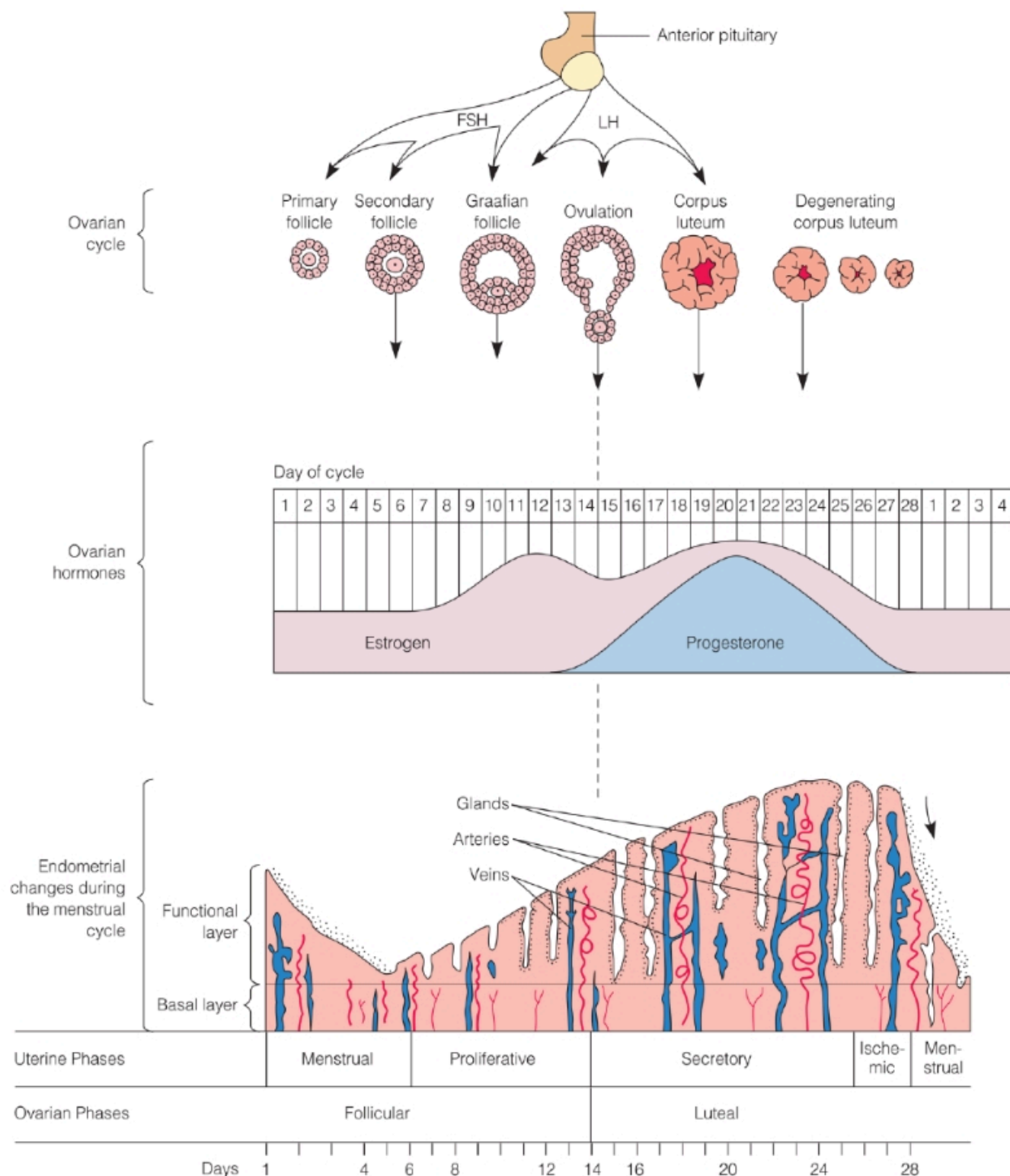
Ovulation occurs when a mature ovum is released from the follicle about 14 days before the onset of the next menstrual period. The corpus luteum turns yellow (luteinizing) immediately after ovulation and secretes increasing quantities of progesterone to prepare the uterine lining for a fertilized ovum. Approximately 12 days after ovulation, the corpus luteum degenerates if fertilization has not occurred, and progesterone and estrogen levels decrease. The drop in estrogen and progesterone levels causes the endometrium to break down, resulting in menstruation.

The anterior pituitary gland secretes more FSH and LH, beginning a new cycle. The climacteric is a period of years during which the woman's ability to reproduce gradually declines.

Menopause refers to the final menstrual period, although in casual use the terms menopause and climacteric are often used interchangeably.

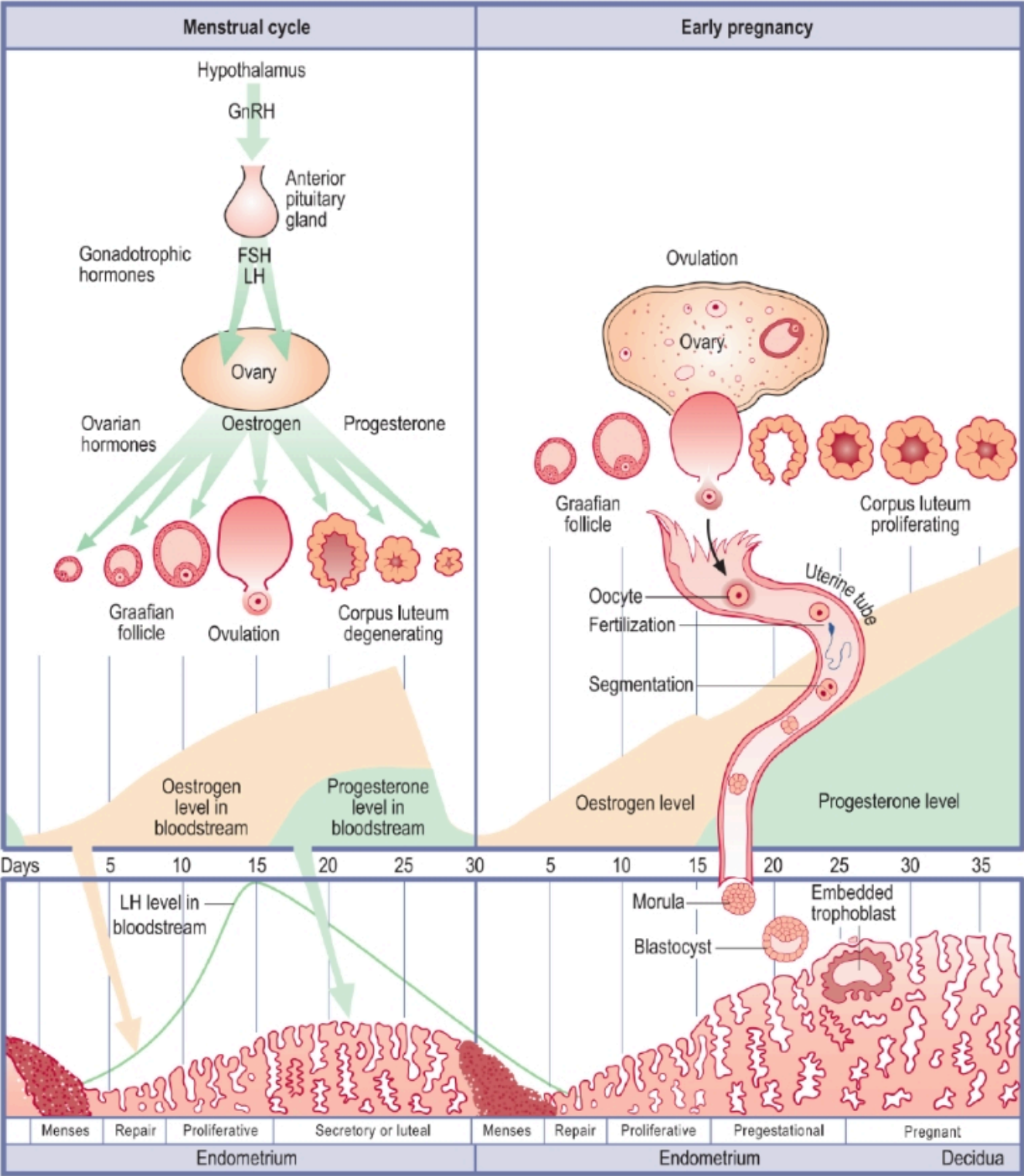


The study of the female reproductive organs is **Gynecology**, the study of male reproductive organs is **Andrology**, and **Obstetrics** is the field of study concentrated on pregnancy, childbirth and the postpartum period.



Female reproductive cycle: interrelationships of hormones with the four phases of the uterine cycle and the two phases of the ovarian cycle in an ideal 28-day cycle.





## References:

- Berek, J. S. (2020). *Berek & Novak's Gynecology Sixteenth Edition*.
- Bergsjø, P. (2001). Clinical Protocols in Obstetrics and Gynecology. The 'TAN' Book. *Acta Obstetrica et Gynecologica Scandinavica*, 80(10), 977–977. <https://doi.org/10.1111/j.1600-0412.2001.801022.pp.x>
- Bradley, R. J. (2000). Lecture Notes on Obstetrics and Gynaecology. In *The Obstetrician & Gynaecologist* (Vol. 2, Issue 2). <https://doi.org/10.1576/toag.2000.2.2.53>
- Care, N. (n.d.). Maternal and Fetal Evaluation and Immediate Newborn Care.
- Chan, F. Y., & Oats, J. J. N. (2002). Obstetrics and gynaecology. *Medical Journal of Australia*, 176(1), 28. <https://doi.org/10.5694/j.1326-5377.2002.tb04261.x>
- Disaia, P. J., Creasman, W. T., & Mannel, Robert S, et al. (2018). *Clinical Gynecology Oncology*. Edition, S. (n.d.). MATERNAL-NEWBORN NURSING.
- Gluckman, P., Hanson, M., Seng, C. Y., & Bardsley, A. (2014). Nutrition and Lifestyle for Pregnancy and Breastfeeding. 27, 432. <https://books.google.com/books?id=qkaPBQAAQBAJ&pgis=1>
- Herdiana. (2013). The science of pregnancy. In *Journal of Chemical Information and Modeling* (Vol. 53, Issue 9).
- Hockenberry, M. J., & David, W. (2015). *Introduction to Maternity and Pediatric Nursing: Study Guide for Leifer, 8th Edition*. 384.
- Horsager, R., Roberts, S. W., Rogers, V. L., Santiago-Muñoz, P. C., Worley, K. C., & Hoffman, B. L. (2014). *William Obstetric Study Guide*.
- Jayne Klossner, & Hatfield, N. (2010). *Introductory Maternity & Pediatric*.
- Johnson, Joyce, Y. (2014). *Maternal - Newborn Nursing Demystified*.
- Leifer, G. (2019). *Introduction to Maternity and Pediatric Nursing: 8th Edition*. Elsevier, 1–1756.
- MICOG Professor, F. (2007). *OBSTETRICS AND GYNECOLOGY* Editor Kanan Yelikar Foreword Nagnath Kottapalle.
- Obstetric in Focus.Pdf. (n.d.).
- Stone, J., Eddleman, K., & Duenwald, M. (2009). *Pregnancy for dummies*. In --For dummies.
- Szymanski LM, B. J. (2016). *Johns Hopkins Handbook of Obstetrics and Gynecology*. In Mc Graw Hill Education.
- W. Ladewig Patricia. (2010). *Contemporary Maternal-Newborn Nursing Care*.
- Webster, S. N. (2009). Creasy & Resnik's *Maternal- Fetal Medicine: Principles and Practice*, 6th edition. In *The Obstetrician & Gynaecologist* (Vol. 11, Issue 4). <https://doi.org/10.1576/toag.11.4.294b.27538>